CUBE PAINTING AND CUTTING

Counting Small Cubes by Number of Painted Faces

OAVS TGT | Reasoning (Additional topic from the “etc.” in the syllabus) | Topic 28

Syllabus line (Reasoning, 30 Marks): The verbal and non-verbal topics listed in the syllabus end with “etc.”. This topic is not named in that line, but it is asked in the papers: a painted-cube cutting question appears in English 2022 (Q40).

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). One cube-cutting question appears (English 2022 Q40). It is solved below. The practice questions were generated or checked by program so that each has exactly one correct option.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The one PYQ and what it asks
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4	Shortcuts, Traps and Scope	Speed methods and mistakes
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CHAPTER 1

PYQ Analysis

One question, in English 2022

Paper	Q.	Question	Type	Ans
English 2022	40	All opposite faces of a big cube are painted yellow, green and purple; cut into 64 equal small cubes; how many have three faces coloured?	Painted cube	C (8)
English 2019 / Mathematics 2024 / Science 2024 / Social Studies 2023	–	none		

PYQ INSIGHT: The PYQ cuts a fully painted cube into 64 small cubes (4 × 4 × 4). The small cubes with **three painted faces are the 8 corners**. Every painted-cube question is answered by classifying the small cubes into **corners, edges, face centres and the inside**.

Pattern	PYQs	Priority
Count small cubes with 3, 2, 1 or 0 painted faces (all faces painted)	1	★★★ Must
Cube painted on only some faces	0	★★ Expected
Number of cuts needed to make the small cubes	0	★★ Expected

CHAPTER 2

Foundation

Position classes, formulas and variants

2.1 A cube cut into n × n × n small cubes (all faces painted)

Small cube position	Painted faces	Count
Corner	3	8
Edge (not corner)	2	$12(n-2)$
Face centre	1	$6(n-2)^2$
Inside	0	$(n-2)^3$

n	2	3	4	5	6
3 faces	8	8	8	8	8
2 faces	0	12	24	36	48
1 face	0	6	24	54	96
0 faces	0	1	8	27	64
Total n^3	8	27	64	125	216

NOTE: The number of cubes with 3 painted faces is always 8 (corners) for $n \geq 2$. The total of the four counts must equal n^3: for n = 4, 8 + 24 + 24 + 8 = 64.

2.2 Variants

Variant	How to count
Some faces not painted	Count by position: a small cube has a painted face for each painted big face it touches
Opposite faces painted differently (the PYQ)	Colours do not change the counts: three painted faces means a corner
Number of cuts to make n^3 cubes	$3(n-1)$ cuts if no stacking is allowed (n = 4: 9 cuts)
Cuts needed for 64 cubes in 3 directions	3 cuts in each of 3 directions

SHORTCUT: Remember **8, $12(n-2)$, $6(n-2)^2$, $(n-2)^3$**. English 2022 Q40: n = 4 ($64 = 4^3$), so three-coloured cubes = 8.

CHAPTER 3

Solved PYQs

English 2022 Q40 with step-by-step reasoning

English 2022 Q40

All the opposite faces of a big cube are painted with yellow, green and purple colours respectively. After that, the cube is cut into 64 small equal cubes. How many small cubes have three faces coloured? (A) 16 (B) 24 (C) 8 (D) 12

1. 64 = 4 × 4 × 4, so each edge is cut into 4 parts (n = 4).
2. All six faces of the big cube are painted (three colours on three pairs of opposite faces).
3. A small cube has three painted faces only if it lies at a **corner** of the big cube, where three painted faces meet.
4. A cube has 8 corners, so 8 small cubes have three coloured faces.
5. The other options are 24 (edge or face-centre cubes), 12 and 16 (not the corner count).

Answer: Option C (8)

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Corner cubes are always 8.**
- **Two faces:** edge cubes, $12(n-2)$.
- **One face:** $6(n-2)^2$.
- **Zero faces:** the inner cube, $(n-2)^3$.
- **Check:** the four counts add up to n^3.
- **Time:** 30 seconds per question.

Trap	How to avoid it
Counting corners for two faces	Corners have three painted faces.
Using 4 for n = 64	64 small cubes means n = 4 ($n^3 = 64$).
Painted colours mattering	Colours do not change the counts, only which faces are painted.
Forgetting unpainted faces	If a face is not painted, do not count it.

Scope

Related PYQs / topics	Where
Dice (opposite faces of a die, cube nets)	Dice (earlier topic)
Paper folding	Paper Folding (Topic 18)

CHAPTER 5

Practice Set

6 original questions; answers on the next page

A. All faces painted

Q1. A cube painted on all faces is cut into 27 equal small cubes. How many have exactly two painted faces?

A. 8 B. 6 C. 12 D. 1

Q2. A cube painted on all faces is cut into 125 equal small cubes. How many have exactly one painted face?

A. 54 B. 36 C. 27 D. 48

Q3. A cube painted on all faces is cut into 64 equal small cubes. How many have no painted face?

A. 4 B. 16 C. 24 D. 8

B. Some faces painted

Q4. A cube is painted on all faces except the bottom, and cut into 27 small cubes. How many small cubes have exactly one painted face?

A. 5 B. 9 C. 8 D. 12

Q5. A cube is painted on its top and bottom faces only, and cut into 64 small cubes. How many small cubes have exactly one painted face?

A. 16 B. 8 C. 32 D. 48

Q6. How many cuts are needed to cut a solid cube into 64 equal small cubes (no stacking)?

A. 9 B. 6 C. 12 D. 16

Answer key

Q	Answer	Reason
1	C	$n = 3$: edge cubes $= 12(n - 2) = 12$.
2	A	$n = 5$: $6(n - 2)^2 = 6 \times 9 = 54$.
3	D	$n = 4$: $(n - 2)^3 = 2^3 = 8$.
4	B	Top-face centre: 1. Centres of the four side faces (middle layer): 4. Bottom-layer cubes in the middle of each bottom edge touch one painted side face and the unpainted bottom: 4. Total $1 + 4 + 4 = 9$ (checked by program).
5	C	Top layer: 16 small cubes with the top painted; bottom layer: 16. No small cube touches both faces, so exactly one painted face for $16 + 16 = 32$ cubes.
6	A	Cut 3 times in each of 3 directions: $3 \times 3 = 9$.

CHAPTER 6

Quick Revision

One page for the last day

Painted faces	Count (n × n × n, all faces painted)
3	8
2	$12(n-2)$
1	$6(n-2)^2$
0	$(n-2)^3$
Total	n^3
Cuts for n^3 cubes	$3(n-1)$

PYQ RULE: English 2022 Q40: $64 = 4^3$, all faces painted, so three-coloured cubes are the 8 corners.

ROUTINE: Exam-day routine: find n from the number of small cubes; note which faces are painted; classify the small cubes; check the total. 30 seconds per question.